CASE STUDY DOCUMENT

Online Appointment Booking System

MediBook

Book Smarter. Heal Faster. Care Better.

Version 1.0 | 2026

1. Synopsis

MediBook is a full-stack Online Appointment Booking System that enables patients to search for healthcare providers (doctors, consultants, therapists, and specialists), view their available time slots, and book appointments online — all from a single, unified platform. The system supports three primary roles: Patients, Providers (healthcare professionals), and Administrators, each with a role-specific dashboard and a distinct set of capabilities.

Patients can browse and search providers by specialization, name, or location; view real-time availability calendars; book, reschedule, or cancel appointments; make secure online payments; and access their medical records and prescriptions after each consultation. Providers can configure their weekly availability slots, manage their appointment queue, mark consultations as complete, create electronic medical records with diagnoses and prescriptions, and track their earnings analytics. Administrators oversee the platform: verifying providers, moderating reviews, managing all users, viewing platform-wide analytics, and handling financial reconciliation.

MediBook is architected as a microservices system modelled on the EShoppingZone reference pattern, with independent services for authentication/user management, provider profiles, availability/scheduling, appointment lifecycle management, payments, reviews/ratings, notifications, and medical records — all integrated through a Spring MVC website controller layer.

2. Detailed Requirements

2.1 General Requirements

- Guests can browse providers and view profiles and available slots without logging in.
- Users must register or log in to book appointments, make payments, or access records.
- Role-based access control with three distinct roles: Patient, Provider, and Admin.
- All users can update their profile details, contact information, and password.
- In-app, email, and SMS notifications are sent for all key booking lifecycle events.
- Admin panel provides full platform management, provider verification, and analytics.

2.2 Patient Requirements

- Register and log in via email/password or OAuth (Google/GitHub).
- Browse the provider directory and filter by specialization, location, rating, or availability.
- View detailed provider profiles including qualifications, experience, clinic address, and average rating.
- View a provider's available time slots for any selected date in real time.
- Book an appointment by selecting a slot and specifying service type and mode (in-person or teleconsultation).
- Cancel an appointment; receive a refund if cancelled within the allowed cancellation window.

- Reschedule an appointment to another available slot for the same provider.
- Make payments online (wallet, card, UPI) or opt for pay-at-clinic.
- View complete appointment history with status (Scheduled, Completed, Cancelled, No-Show).
- View electronic medical records and prescriptions created by the provider after each visit.
- Submit a star rating and written review for a provider after a completed appointment.
- Receive in-app and email/SMS notifications for: booking confirmation, reminders (24h and 1h before), cancellation alerts, and payment receipts.

2.3 Provider Requirements

- Register as a provider and submit qualification and specialization details for admin verification.
- Configure weekly availability by adding individual or recurring time slots.
- Block specific slots for personal unavailability or leave periods.
- View today's appointment schedule and full upcoming appointment list.
- Mark appointments as completed, marking the slot as done and unlocking the review capability for the patient.
- Create electronic medical records after each appointment — including diagnosis, prescription, and follow-up date.
- Edit medical records within an allowed editing window after the appointment.
- View earnings analytics: total revenue, pending payments, and monthly breakdowns.
- View all reviews and ratings given by patients; flag inappropriate reviews for admin moderation.
- Receive notifications for: new bookings, cancellations, rescheduling requests, and follow-up reminders.

2.4 Admin Requirements

- View and manage all user accounts: patients and providers — suspend, reactivate, or delete.
- Review new provider registrations and verify or reject their credentials.
- View all appointments across the platform with full lifecycle status.
- View all payment transactions; trigger refunds where applicable.
- View and moderate all provider reviews; remove inappropriate or fraudulent reviews.
- Access platform analytics: total bookings, revenue, most-booked specializations, completion rates.
- Send platform-wide notifications to patients, providers, or all users.
- Generate revenue reports for billing and financial reconciliation.
- View all medical records for audit compliance (read-only access).

2.5 Availability & Scheduling Requirements

- Providers define slots with a specific date, start time, end time, and duration in minutes.
- Recurring slot generation supports daily, weekly, and custom patterns across a date range.

- A slot's status transitions: Available → Booked on appointment creation; Released on cancellation.
- Blocked slots are invisible to patients but visible to the provider and admin.
- Expired slots (past date/time with no booking) are automatically purged by a scheduled job.

2.6 Payment Requirements

- Patients can pay online at booking time or at the clinic after the appointment.
- Online payment modes: wallet balance, debit/credit card, UPI, or net banking.
- Each payment record stores amount, mode, transaction ID, timestamp, and status.
- Refunds are processed within 3–5 business days for eligible cancellations.
- Providers receive payment summaries (total collected, pending, refunded) in their earnings dashboard.

2.7 Medical Record Requirements

- Records are created by the provider after marking an appointment complete.
- Each record stores diagnosis, prescription, clinical notes, and optional document attachment URL.
- Patients can view their own records; providers can view records they created; admins have read-only audit access.
- Records with a follow-up date trigger an automated reminder notification to the patient on that date.

3. Use Case Diagram

The diagram below captures all actors and use cases within the MediBook platform — spanning authentication, provider browsing, appointment booking and lifecycle management, payments, medical records, reviews, notifications, and administrative operations.

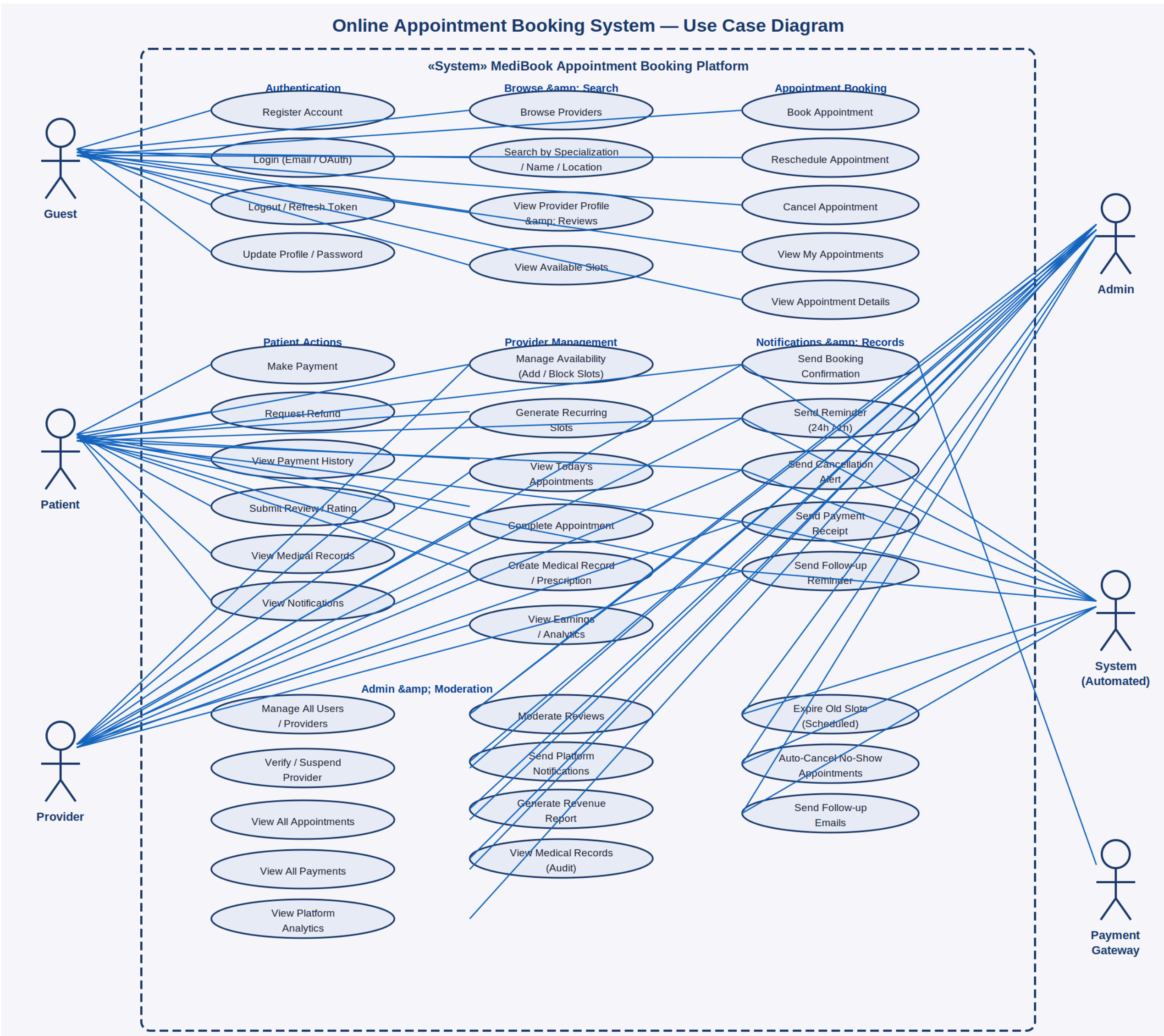

Figure 1: MediBook Online Appointment Booking System — Complete Use Case Diagram

3.1 Actors

Actor	Description
Guest	Unauthenticated visitor who can browse providers, view profiles, and check available slots
Patient	Registered user who books appointments, makes payments, views records, and submits reviews
Provider	Healthcare professional who manages availability, conducts appointments, and creates medical records
Admin	Platform administrator who manages users, verifies providers, moderates content, and views analytics
System (Automated)	Automated component handling notifications, slot expiry, no-show detection, and follow-up reminders
Payment Gateway	External service processing online appointment payment and refund transactions

3.2 Use Case Summary

Use Case	Actor(s)	Description
Register Account	Guest	Create a patient or provider account with email/password or OAuth
Login / Logout	Guest, Patient, Provider	Authenticate or end a session; refresh JWT token
Update Profile	Patient, Provider	Change name, phone, profile picture, or password
Browse Providers	Guest, Patient	Browse the provider directory with filter options
Search by Specialization	Guest, Patient	Find providers by specialization, name, or clinic location
View Provider Profile	Guest, Patient	See qualifications, bio, clinic info, ratings, and reviews
View Available Slots	Guest, Patient	Check real-time slot availability for a selected provider and date
Book Appointment	Patient	Select a slot and confirm booking with service type and mode
Reschedule Appointment	Patient	Move booking to another available slot for the same provider
Cancel Appointment	Patient, Provider	Cancel a booking; trigger refund if within cancellation policy
View My Appointments	Patient	List all past and upcoming appointments with status

Use Case	Actor(s)	Description
Make Payment	Patient, Payment Gateway	Pay for appointment online via card, UPI, or wallet
Request Refund	Patient	Submit a refund request for a cancelled or missed appointment
View Payment History	Patient	View all transaction records with amounts and statuses
Submit Review / Rating	Patient	Rate a provider (1-5 stars) and write a review after consultation
View Medical Records	Patient	Access electronic records and prescriptions from past consultations
Manage Availability	Provider	Add, edit, block, or delete availability slots
Generate Recurring Slots	Provider	Auto-create slots for a date range using a recurrence pattern
View Today's Appointments	Provider	See the day's appointment schedule in chronological order
Complete Appointment	Provider	Mark a consultation as done and unlock record creation
Create Medical Record	Provider	Write diagnosis, prescription, notes, and set follow-up date
View Earnings	Provider	View revenue breakdown by period with pending and received amounts
Verify Provider	Admin	Review and approve or reject a provider's credential submission
Manage All Users	Admin	View, suspend, reactivate, or delete any patient or provider account
View All Appointments	Admin	Monitor all bookings platform-wide with full status details
View Platform Analytics	Admin	Access total bookings, revenue, completion rates, and top specializations
Moderate Reviews	Admin	Remove inappropriate or fraudulent patient reviews
Generate Revenue Report	Admin	Export financial summaries for billing and reconciliation
Send Booking Confirmation	System	Auto-dispatch booking confirmation email/SMS to patient and provider
Send Reminder	System	Send 24-hour and 1-hour pre-appointment reminders

Use Case	Actor(s)	Description
Auto-Cancel No-Show	System	Flag appointments as No-Show if not completed within a time window
Send Follow-up Reminder	System	Notify patient on follow-up date set in the medical record

4. Class Diagrams — All Services

MediBook follows the same layered microservices architecture as the EShoppingZone reference system. Each service comprises five layers: Entity/POJO (domain model), Repository Interface (data access), Service Interface (business contract), ServiceImpl (business logic), and REST Resource (API exposure). The ten class diagrams below detail every service in the platform.

4.1 Auth / User-Service

The Auth/User-Service is the security gateway for the entire platform. It manages registration, login, JWT lifecycle, OAuth2 provider integration (Google/GitHub), profile management, password changes, and account deactivation. Every protected resource on the platform is gated through this service's token validation logic.

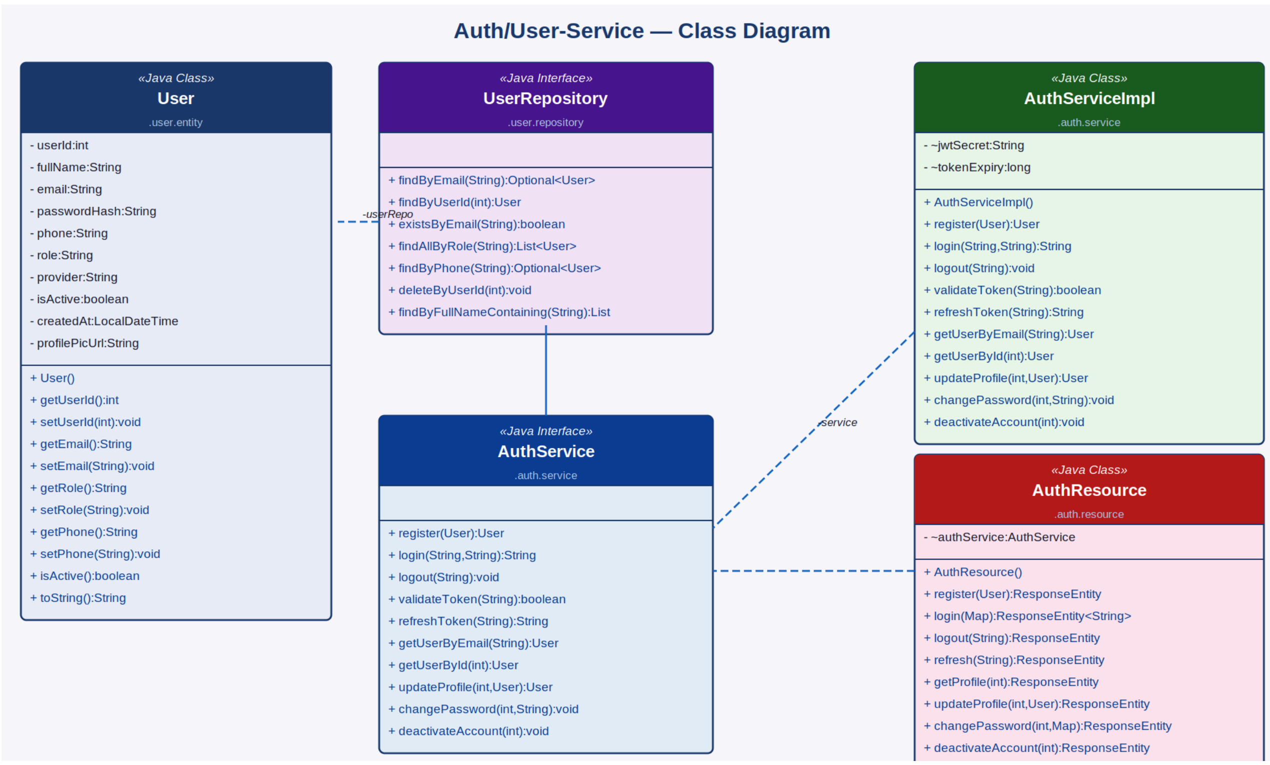

Figure 2: Auth/User-Service — Class Diagram

Component	Type	Responsibility
User	Java Class (Entity)	Stores userId, fullName, email, passwordHash, phone, role (Patient/Provider/Admin), provider (OAuth), isActive, createdAt, profilePicUrl
UserRepository	Java Interface	findByEmail(), findByUserId(), existsByEmail(), findAllByRole(), findByPhone(), findByFullNameContaining(), deleteByUserId()
AuthServiceImpl	Java Class	Implements registration, login, JWT generation/validation, OAuth2 handling, profile update, password change, and account deactivation
AuthService	Java Interface	Declares register(), login(), logout(), validateToken(), refreshToken(), getUserByEmail(), getUserById(), updateProfile(), changePassword(), deactivateAccount()

Component	Type	Responsibility
AuthResource	Java Class (REST)	Exposes /auth/register, /auth/login, /auth/logout, /auth/refresh, /auth/profile (GET/PUT), /auth/password, /auth/deactivate endpoints

4.2 Provider-Service

The Provider-Service manages professional profiles of healthcare providers on the platform. It stores specialization, qualifications, experience, clinic details, and aggregated rating. Admin verification is required before a provider appears in patient search results. The service supports full-text search by name, specialization, or location and exposes availability and verified-status flags.

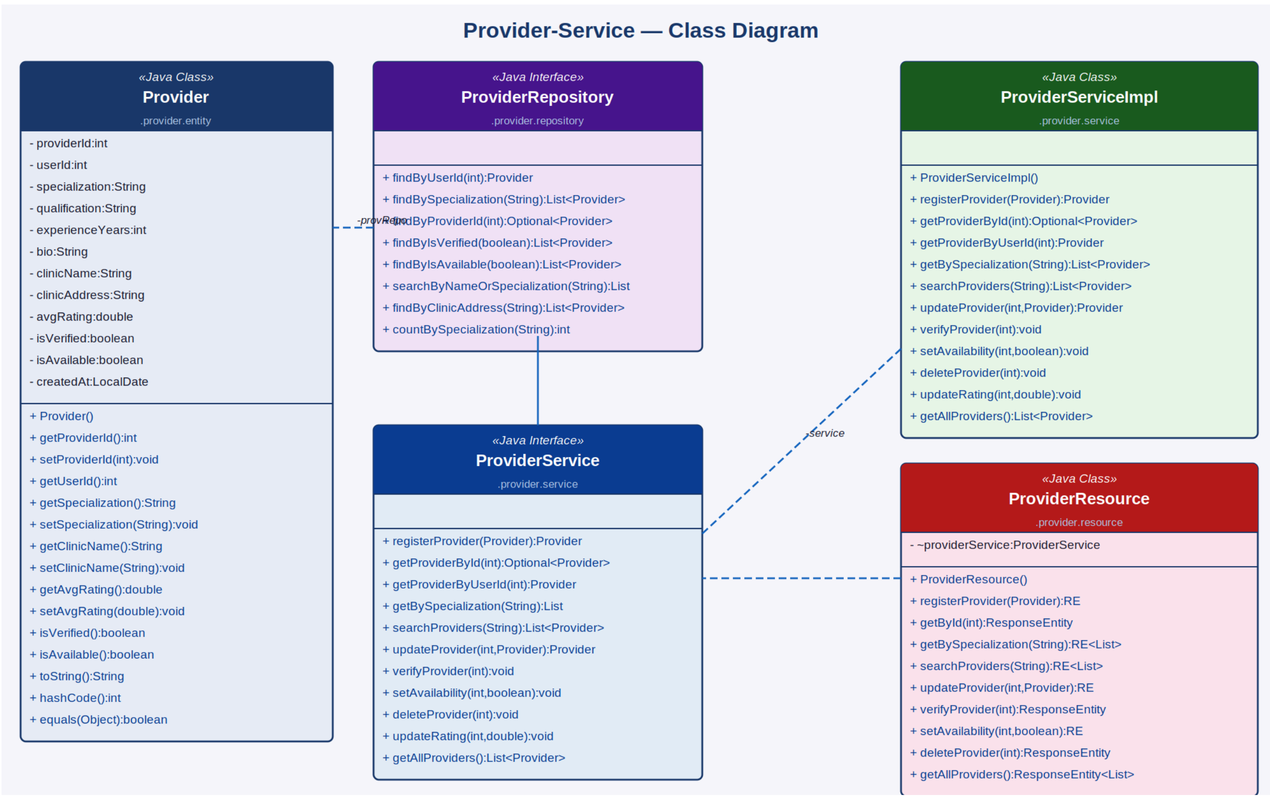

Figure 3: Provider-Service — Class Diagram

Component	Type	Key Attributes / Methods
Provider	Java Class (Entity)	providerId, userId, specialization, qualification, experienceYears, bio, clinicName, clinicAddress, avgRating, isVerified, isAvailable, createdAt
ProviderRepository	Java Interface	findByUserId(), findBySpecialization(), findByIsVerified(), findByIsAvailable(), searchByNameOrSpecialization(), findByClinicAddress(), countBySpecialization()
ProviderServiceImpl	Java Class	registerProvider(), getProviderById(), getBySpecialization(), searchProviders(), updateProvider(), verifyProvider(), setAvailability(), deleteProvider(), updateRating(), getAllProviders()
ProviderService	Java Interface	Declares all provider CRUD, search, verification, availability, and rating operations

Component	Type	Key Attributes / Methods
ProviderResource	Java Class (REST)	Exposes /providers endpoints: POST (register), GET (by id/specialization/search/all), PUT (update/verify/availability), DELETE

4.3 Availability / Schedule-Service

The Availability/Schedule-Service manages provider time slots. Each AvailabilitySlot records a provider's date, start/end time, duration, and booking status. Providers can add individual slots, bulk-create them, block specific slots for leave, or generate recurring slots using a daily or weekly pattern. The service enforces slot transitions (Available → Booked → Released) and exposes only unblocked, unbooked slots to patients.

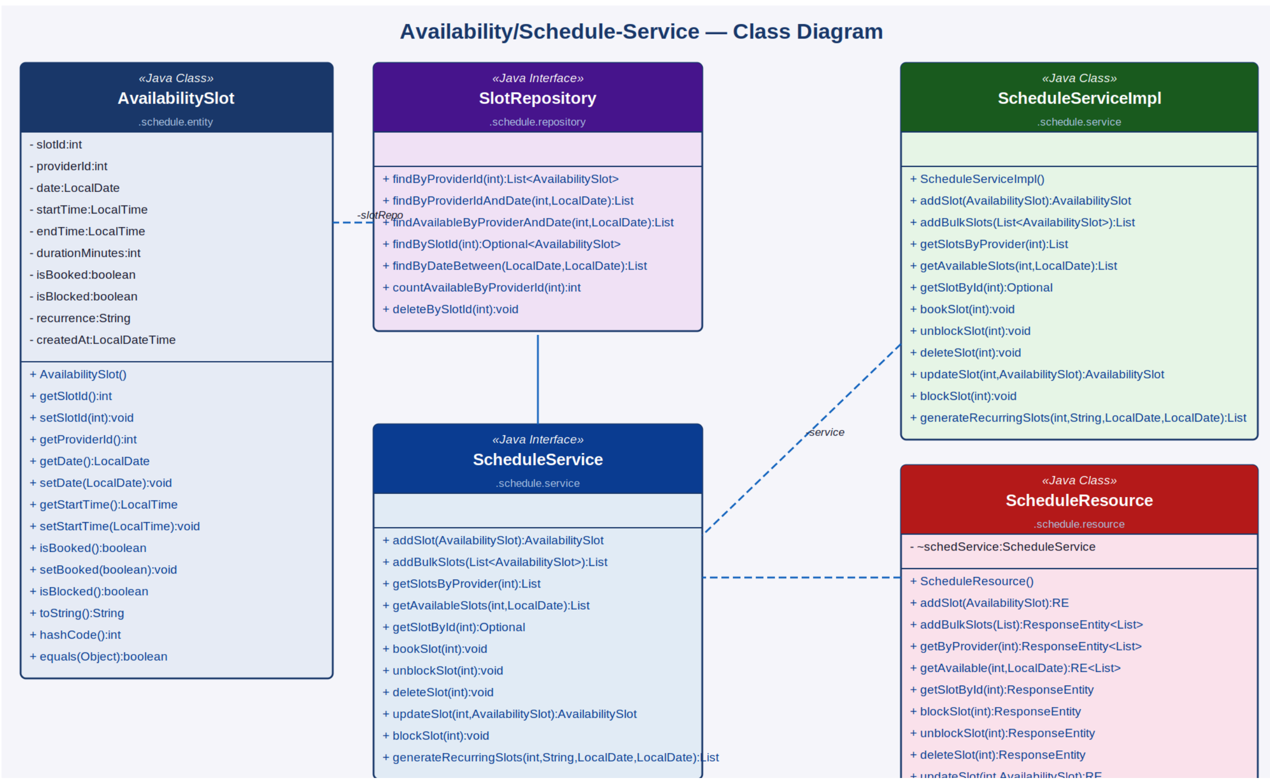

Figure 4: Availability/Schedule-Service — Class Diagram

Component	Type	Key Attributes / Methods
AvailabilitySlot	Java Class (Entity)	slotId, providerId, date, startTime, endTime, durationMinutes, isBooked, isBlocked, recurrence, createdAt
SlotRepository	Java Interface	findByProviderId(), findByProviderIdAndDate(), findAvailableByProviderAndDate(), findByDateBetween(), countAvailableByProviderId(), deleteBySlotId()
ScheduleServiceImpl	Java Class	addSlot(), addBulkSlots(), getSlotsByProvider(), getAvailableSlots(), getSlotById(), bookSlot(), unblockSlot(), deleteSlot(), updateSlot(), blockSlot(), generateRecurringSlots()

Component	Type	Key Attributes / Methods
ScheduleService	Java Interface	Declares all slot CRUD, booking state management, bulk creation, and recurrence generation operations
ScheduleResource	Java Class (REST)	Exposes /slots endpoints: POST (add/bulk/generateRecurring), GET (by provider/available/id), PUT (block/unblock/update), DELETE

4.4 Appointment-Service

The Appointment-Service is the core orchestration service of the platform. It manages the full booking lifecycle — creation, rescheduling, cancellation, and completion. Each Appointment links a patient to a provider's slot and records the service type, consultation mode (in-person or teleconsultation), clinical notes, and status. On booking, it calls the Schedule-Service to mark the slot as booked; on cancellation, it releases the slot and triggers a refund via the Payment-Service.

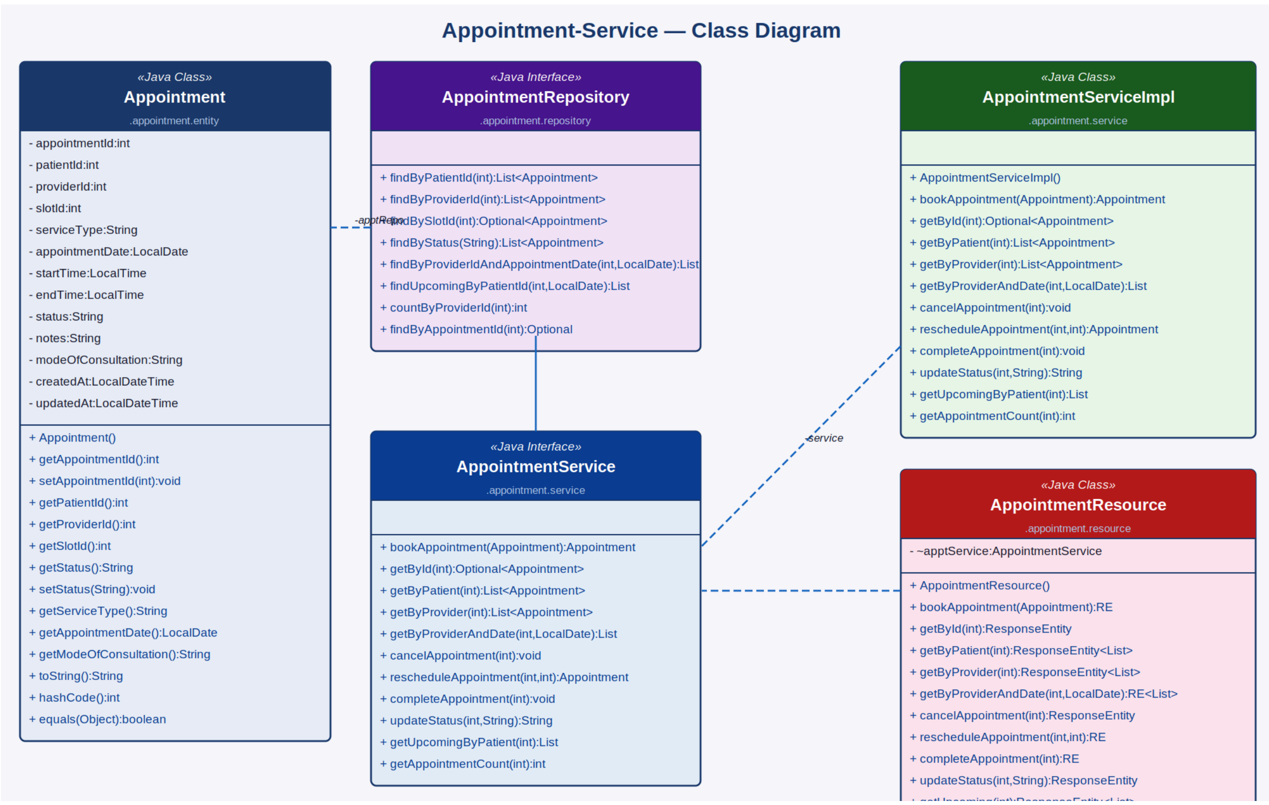

Figure 5: Appointment-Service — Class Diagram

Component	Type	Key Attributes / Methods
Appointment	Java Class (Entity)	appointmentId, patientId, providerId, slotId, serviceType, appointmentDate, startTime, endTime, status (Scheduled/Completed/Cancelled/No-Show), notes, modeOfConsultation, createdAt, updatedAt
AppointmentRepository	Java Interface	findByPatientId(), findByProviderId(), findBySlotId(), findByStatus(), findByProviderIdAndAppointmentDate(), findUpcomingByPatientId(), countByProviderId()
AppointmentServiceImpl	Java Class	bookAppointment(), getById(), getByPatient(), getByProvider(), getByProviderAndDate(), cancelAppointment(), rescheduleAppointment(),

Component	Type	Key Attributes / Methods
		completeAppointment(), updateStatus(), getUpcomingByPatient(), getAppointmentCount()
AppointmentService	Java Interface	Declares all appointment booking, retrieval, lifecycle transition, and count operations
AppointmentResource	Java Class (REST)	Exposes /appointments endpoints: POST (book), GET (by id/patient/provider/date/upcoming), PUT (cancel/reschedule/complete/status), GET count

4.5 Payment-Service

The Payment-Service processes appointment payments and manages refunds. Each Payment record is linked to exactly one appointment and stores the amount, payment mode, transaction ID, currency, and status. The service supports invoice generation for completed appointments and exposes earnings aggregation for provider revenue dashboards. Refunds are triggered on appointment cancellation and processed through the payment gateway.

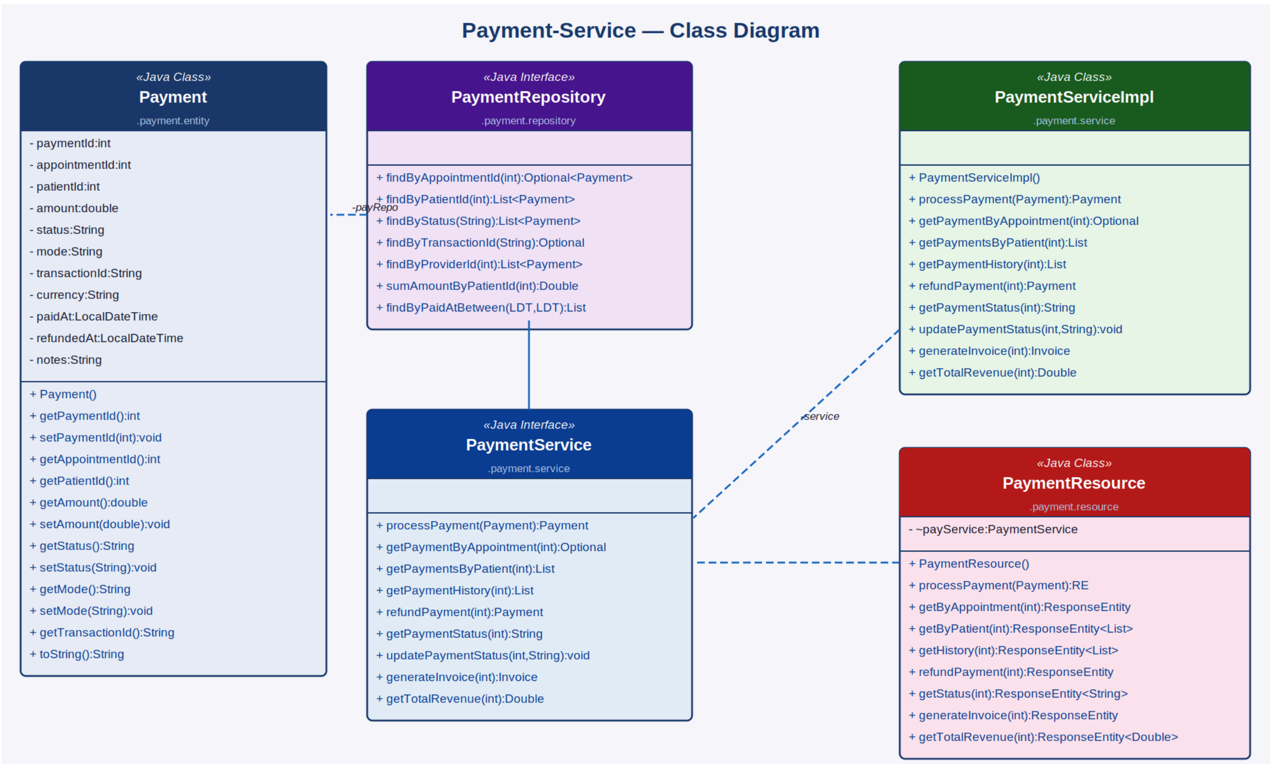

Figure 6: Payment-Service — Class Diagram

Component	Type	Key Attributes / Methods
Payment	Java Class (Entity)	paymentId, appointmentId, patientId, amount, status (Pending/Paid/Refunded/Failed), mode (Card/UPI/Wallet/Cash), transactionId, currency, paidAt, refundedAt, notes
PaymentRepository	Java Interface	findByAppointmentId(), findByPatientId(), findByStatus(), findByTransactionId(), findByProviderId(), sumAmountByPatientId(), findByPaidAtBetween()
PaymentServiceImpl	Java Class	processPayment(), getPaymentByAppointment(), getPaymentsByPatient(), getPaymentHistory(), refundPayment(), getPaymentStatus(), updatePaymentStatus(), generateInvoice(), getTotalRevenue()

Component	Type	Key Attributes / Methods
PaymentService	Java Interface	Declares all payment processing, refund, invoice generation, and revenue aggregation operations
PaymentResource	Java Class (REST)	Exposes /payments endpoints: POST (process), GET (by appointment/patient/history), POST (refund), GET (status, invoice, totalRevenue)

4.6 Review / Rating-Service

The Review/Rating-Service allows patients to rate and review providers after a completed appointment. One review is permitted per appointment, enforced by a unique constraint on appointmentId. The service computes the average rating per provider (used by the Provider-Service to update the avgRating field) and supports anonymous reviews. Admin moderation can remove inappropriate content.

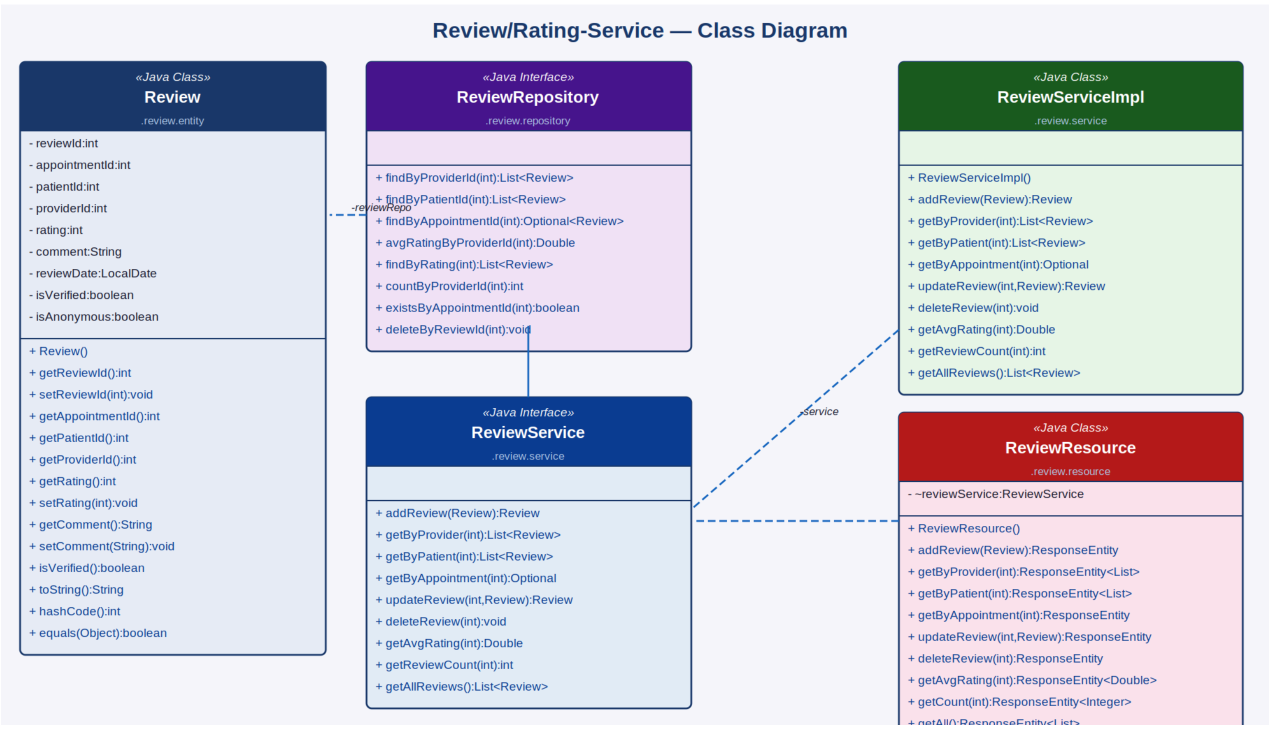

Figure 7: Review/Rating-Service — Class Diagram

Component	Type	Key Attributes / Methods
Review	Java Class (Entity)	reviewId, appointmentId (unique), patientId, providerId, rating (1-5), comment, reviewDate, isVerified, isAnonymous
ReviewRepository	Java Interface	findByProviderId(), findByPatientId(), findByAppointmentId(), avgRatingByProviderId(), findByRating(), countByProviderId(), existsByAppointmentId(), deleteByReviewId()
ReviewServiceImpl	Java Class	addReview(), getByProvider(), getByPatient(), getByAppointment(), updateReview(), deleteReview(), getAvgRating(), getReviewCount(), getAllReviews()
ReviewService	Java Interface	Declares all review CRUD, average rating computation, and moderation operations

Component	Type	Key Attributes / Methods
ReviewResource	Java Class (REST)	Exposes /reviews endpoints: POST (add), GET (by provider/patient/appointment/all), PUT (update), DELETE, GET (avgRating, count)

4.7 Notification-Service

The Notification-Service dispatches in-app, email, and SMS alerts for all booking lifecycle events: booking confirmation, 24-hour and 1-hour reminders, cancellation alerts, payment receipts, and follow-up reminders. Notifications carry a relatedId and relatedType for deep-linking from the notification centre. The service tracks read/unread state and exposes an unread badge count. Admin can broadcast platform-wide messages via the bulk dispatch endpoint.

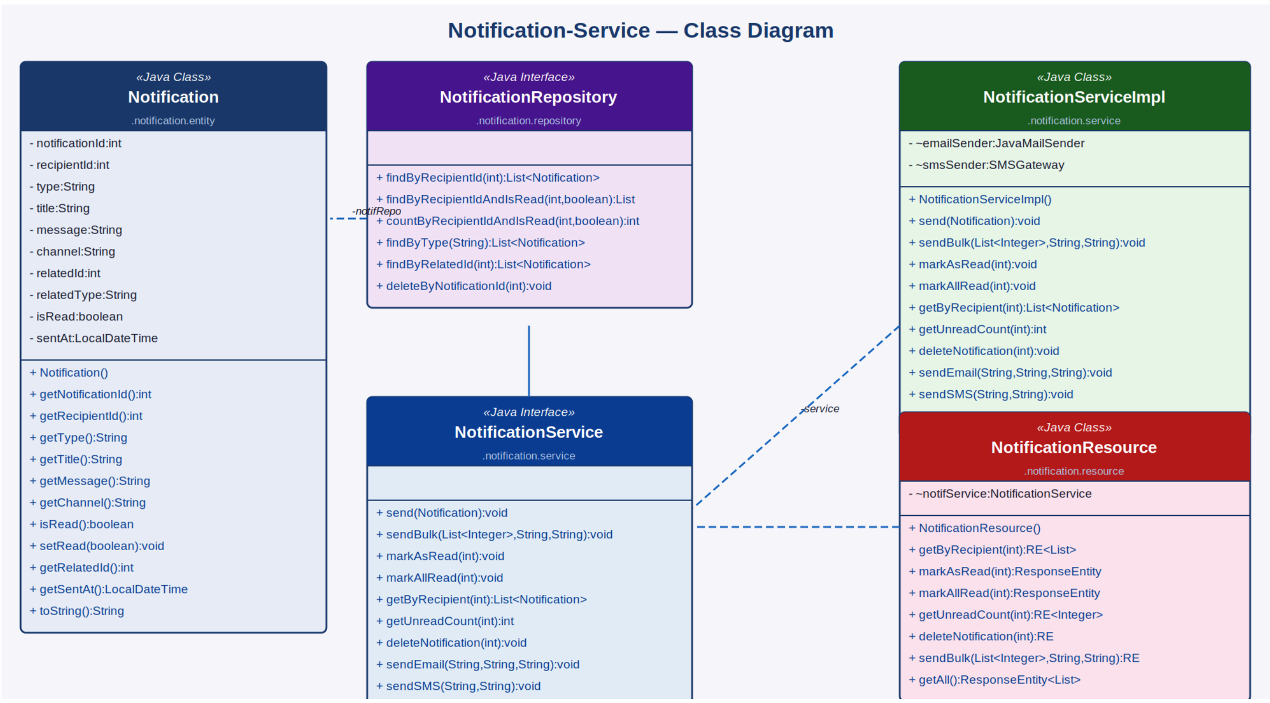

Figure 8: Notification-Service — Class Diagram

Component	Type	Key Attributes / Methods
Notification	Java Class (Entity)	notificationId, recipientId, type (BOOKING/REMINDER/CANCELLATION/PAYMENT/FOLLOWUP), title, message, channel (APP/EMAIL/SMS), relatedId, relatedType, isRead, sentAt
NotificationRepository	Java Interface	findByRecipientId(), findByRecipientIdAndIsRead(), countByRecipientIdAndIsRead(), findByType(), findByRelatedId(), deleteByNotificationId()
NotificationServiceImpl	Java Class	send(), sendBulk(), markAsRead(), markAllRead(), getByRecipient(), getUnreadCount(), deleteNotification(), sendEmail(), sendSMS(), getAll()
NotificationService	Java Interface	Declares notification dispatch (single/bulk/email/SMS), retrieval, read-state management, and delete operations

Component	Type	Key Attributes / Methods
NotificationResource	Java Class (REST)	Exposes /notifications endpoints: getByRecipient, markAsRead, markAllRead, getUnreadCount, delete, sendBulk, getAll

4.8 Medical Record-Service

The Medical Record-Service stores electronic health records created by providers after each completed appointment. Each MedicalRecord is linked to an appointment and contains the diagnosis, prescription, clinical notes, an optional document attachment URL, and a follow-up date. Patients can read their own records; providers access records they created. Records with a follow-up date trigger an automated reminder notification via the Notification-Service.

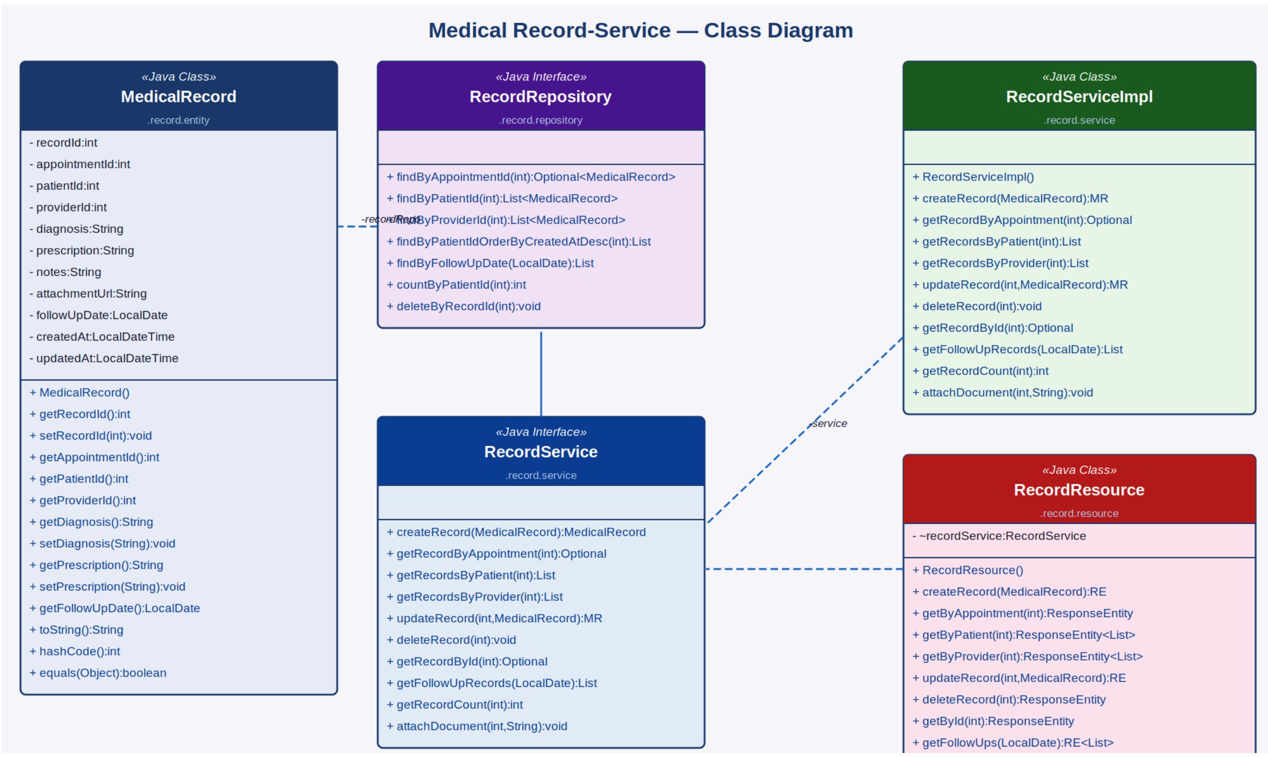

Figure 9: Medical Record-Service — Class Diagram

Component	Type	Key Attributes / Methods
MedicalRecord	Java Class (Entity)	recordId, appointmentId (unique), patientId, providerId, diagnosis, prescription, notes, attachmentUrl, followUpDate, createdAt, updatedAt
RecordRepository	Java Interface	findByAppointmentId(), findByPatientId(), findByProviderId(), findByPatientIdOrderByCreatedAtDesc(), findByFollowUpDate(), countByPatientId(), deleteByRecordId()
RecordServiceImpl	Java Class	createRecord(), getRecordByAppointment(), getRecordsByPatient(), getRecordsByProvider(), updateRecord(), deleteRecord(), getRecordById(), getFollowUpRecords(), getRecordCount(), attachDocument()

Component	Type	Key Attributes / Methods
RecordService	Java Interface	Declares all medical record CRUD, follow-up retrieval, document attachment, and audit access operations
RecordResource	Java Class (REST)	Exposes /records endpoints: POST (create), GET (by appointment/patient/provider/id/followUps), PUT (update/attachDocument), DELETE

4.9 Website-Controller (MVC Layer)

The Website-Controller is the Spring MVC integration layer that wires all underlying microservices together and renders Thymeleaf views for patients, providers, and administrators. It contains three controllers — PatientController, ProviderController, and AdminController — each injecting the relevant service dependencies via Spring @Autowired and delegating to REST Resource endpoints via RestTemplate for inter-service calls.

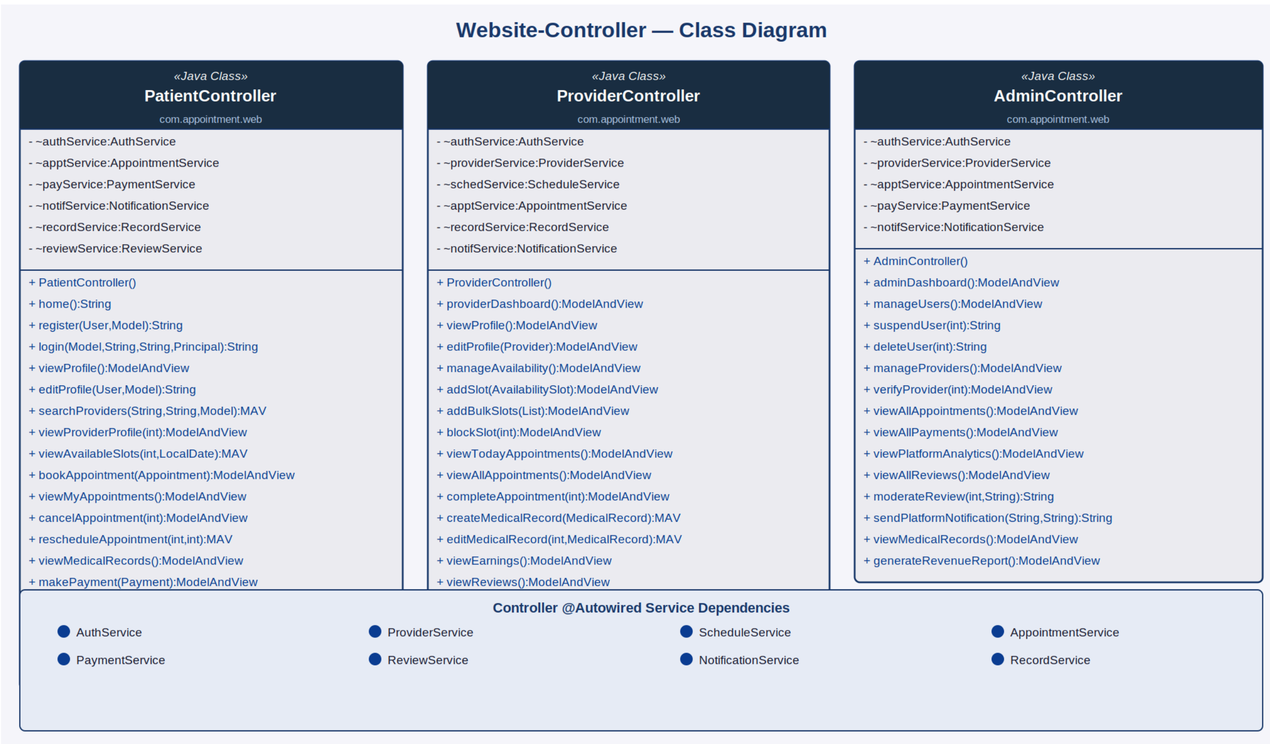

Figure 10: Website-Controller — Class Diagram

Controller	Type	Key Methods
PatientController	Java Class (MVC)	home(), register(), login(), viewProfile(), editProfile(), searchProviders(), viewProviderProfile(), viewAvailableSlots(), bookAppointment(), viewMyAppointments(), cancelAppointment(), rescheduleAppointment(), viewMedicalRecords(), makePayment(), viewPaymentHistory(), submitReview(), viewNotifications(), logout()
ProviderController	Java Class (MVC)	providerDashboard(), viewProfile(), editProfile(), manageAvailability(), addSlot(), addBulkSlots(), blockSlot(), viewTodayAppointments(), viewAllAppointments(), completeAppointment(), createMedicalRecord(), editMedicalRecord(), viewEarnings(), viewReviews(), viewNotifications()

Controller	Type	Key Methods
AdminController	Java Class (MVC)	adminDashboard(), manageUsers(), suspendUser(), deleteUser(), manageProviders(), verifyProvider(), viewAllAppointments(), viewAllPayments(), viewPlatformAnalytics(), viewAllReviews(), moderateReview(), sendPlatformNotification(), viewMedicalRecords(), generateRevenueReport()

5. Microservices Architecture Overview

MediBook follows the same independently deployable microservices pattern as the EShoppingZone reference system. Each service owns its database schema, REST API contract, service interface, and business logic. Synchronous inter-service calls use RestTemplate; asynchronous notification and reminder events are routed via a message broker (RabbitMQ).

Microservice	Base Package	Primary Domain
auth-service	com.medibook.auth	User registration, login, JWT, OAuth2 (Google/GitHub)
provider-service	com.medibook.provider	Provider profiles, search, verification, rating aggregation
schedule-service	com.medibook.schedule	Availability slots, booking state, bulk/recurring slot generation
appointment-service	com.medibook.appointment	Appointment booking, lifecycle management, status transitions
payment-service	com.medibook.payment	Payment processing, refunds, invoices, provider earnings
review-service	com.medibook.review	Patient reviews, star ratings, average rating computation
notification-service	com.medibook.notification	In-app, email, and SMS notifications; bulk dispatch
record-service	com.medibook.record	Electronic medical records, prescriptions, follow-up tracking
medibook-web	com.medibook.web	Spring MVC controllers, Thymeleaf view rendering

6. Non-Functional Requirements

Category	Requirement
Performance	Provider search results and slot availability must respond within 1.5 seconds for 10,000 concurrent users
Scalability	Each microservice independently scalable via Docker/Kubernetes; slot cache in Redis for high-read availability
Security	Passwords stored as bcrypt hashes; JWT expires in 24 hours; OAuth2 for social login; HTTPS enforced; medical records accessible only by patient or their provider
Availability	99.9% uptime SLA; health-check endpoints per service; graceful degradation if one service is temporarily unreachable
Data Integrity	Slot booking is atomic — concurrent booking attempts for the same slot are serialised via optimistic locking to prevent double-booking
HIPAA Compliance	Medical records are encrypted at rest (AES-256) and in transit (TLS 1.3); audit logs are maintained for all record access
Notification SLA	Booking confirmations are dispatched within 30 seconds; appointment reminders are scheduled with second-level precision
Usability	Responsive UI on desktop, tablet, and mobile; WCAG 2.1 AA accessibility compliance
Audit Trail	All appointment status changes, payment transactions, and medical record edits are logged with timestamps and actor IDs
Maintainability	Services independently deployable; RESTful API contracts versioned under /api/v1/

7. Proposed Technology Stack

Layer	Technology
Backend Framework	Java Spring Boot (Spring MVC, Spring Security, Spring Data JPA)
Authentication	JWT (JSON Web Tokens), Spring Security OAuth2, Google / GitHub OAuth
Database	MySQL (primary relational store per service); Redis (slot availability cache and session store)
Calendar / Scheduling	Spring @Scheduled + Quartz Scheduler for slot expiry, reminder dispatch, and no-show detection
File Storage	AWS S3 for medical record document and prescription PDF attachments
Email / SMS	JavaMailSender (SMTP) for email; Twilio or AWS SNS for SMS notifications
Frontend	Thymeleaf templates (Spring MVC) with Bootstrap 5 and FullCalendar.js for the availability calendar UI
Video Consultation	Twilio Video or Jitsi Meet API integration for teleconsultation sessions
Payment Gateway	Razorpay / Stripe for card and UPI payments; internal wallet for prepaid balance
Messaging	RabbitMQ for asynchronous notification dispatch and follow-up reminder scheduling
Containerisation	Docker with Docker Compose; Kubernetes for production orchestration and autoscaling
API Documentation	Swagger / OpenAPI 3.0 for all REST endpoints across all services
CI/CD	GitHub Actions or Jenkins pipeline for automated build, test, and deployment

8. Glossary

Term	Definition
Patient	A registered user who books appointments and consults with healthcare providers on MediBook
Provider	A verified healthcare professional (doctor, therapist, or specialist) who offers appointment slots on the platform
Slot	A defined time block in a provider's calendar during which they are available for a single appointment
Appointment	A confirmed booking linking a patient to a provider's slot with a specific service type and consultation mode
Consultation Mode	The format of the appointment: In-Person (at the clinic) or Teleconsultation (video call)
Service Type	The category of healthcare service requested, e.g., General Consultation, Follow-Up, Dental Checkup
Medical Record	An electronic health record created by the provider after a consultation, containing diagnosis, prescription, and notes
Prescription	A documented list of medications and dosage instructions included in a medical record
Follow-Up Date	A date set by the provider in a medical record triggering an automated patient reminder notification
Recurrence	A pattern (Daily/Weekly) used to auto-generate multiple availability slots across a specified date range
Verification	The admin approval process for a provider's credentials before they appear in patient search results
Refund	A payment reversal triggered when a booking is cancelled within the platform's cancellation policy window
JWT	JSON Web Token — a compact, URL-safe token used for stateless authentication across services
OAuth2	Open Authorization 2.0 — used here for Google and GitHub social login
Microservice	An independently deployable service responsible for a single, well-defined business domain
No-Show	An appointment status assigned automatically when neither party marks it complete within the session window